HHI Report

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Overview

This analysis seeks to evaluate the relationship between an African country's Herfindahl-Hirschman Index (HHI) and their democracy score. The HHI measures the diversity of an African country's official development assistance gross (ODAG) total. Values near 1 indicate a country receiving nearly all of their aid from a single country whereas smaller values of HHI indicate a more balanced ODAG portfolio. We used democracy scores from Freedom House (TOTLIB1), the POLITY IV database (POLITY2), and the Liberal Democracy Index from the V-Dem Institute (V2x_Libdem). (Note: Here we only analyze POLITY2, whereas in previous reports we have analyzed both POLITY and POLITY2.)

This main analyses were split into a reduced and full model. The reduced model only considers democracy as a fixed effect, while including random intercepts for the African country and year. These random effects account for structural correlation between observations within the data structure.

The full model includes all of the parts of the reduced model while adding the African country's population, gross national income (GNI), total trade imports, total trade exports,and total ODAG (Note: When ODAG is not observed it is set to 0 in the analysis). Additionally, we included categorical variables for colonial past and region, and a cold war indicator. We applied a log transformation to GNI, population, total ODAG, imports, and exports due to their right skew. Quadratic and cubic transformations were added when necessary to capture nonlinear relationships. We applied this model framework to current year democracy scores as well as lagged democracy scores of 1, 2, and 3 years.

In the reduced models, every V2x_Libdem and POLITY2 coefficient is significant at $\alpha = 0.05$. The negative direction of their coefficients indicate that, as an African country is more democratic, their HHI drops on average meaning that they tend to have more external countries providing development aid and vice versa. Results for the significance of democracy variables in the reduced model are displayed in Tables 1-3.

In the full models, V2x_Libdem and TOTLIB are significant at the $\alpha = 0.05$ level for all lags. Conversely, POLITY2 is not significant at any lags. The negative coefficient for V2x_Libdem and the positive direction for TOTLIB1 yield the same conclusion as the the reduced models. That is, as an African country is more democratic, it tends to have lower HHI scores on average indicating a greater diversity of countries providing development aid and vice versa. Results for the significance of democracy variables in the reduced model are displayed in Tables 4-6. In addition to the democracy variables, population, GNI, and ODAG are all highly associated with HHI. The Cold War indicator is significant in the POLITY2 and v2x_Libdem models, but not in TOTLIB1 model. The Cold War indicator is positive indicating that during the Cold War HHI values were generally larger (i.e., less diverse) than after the Cold War ended. Finally, all models see socialist ideology during the cold was has a significant negative association. This means that socialist ideology during the cold war is associated with lower HHI (i.e., more diversity) relative to capitalist ideology.

Models

Reduced Model

The reduced model is described below, where the response Y_{it} (Y being HHI) ranges from $1/k$, k being the number of possible sources of ODAG, to 1 for African country i in year t. We applied a log transformation to HHI for better model fit. The predictor X_{1it} represents the democracy score of African country i in year t.

In lagged models, the year t changes units depending on the lag. For instance, a one-year lag would produce the model equation $\log(HHI_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$, where $X_{1i(t-1)}$ is the external country's democracy score in the prior year (year t-1).

$$Y_{it} \sim \mathrm{N}(\mu_{it}, \sigma^2)$$

$$\log(HHI_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$$

$$\gamma_{0i} \sim \mathrm{N}(0, \sigma^2_{country})$$

$$\gamma_{0t} \sim \mathrm{N}(0, \sigma^2_{year})$$

Full Model

Our full model equations are given below. They follow the same setup as the reduced model with the addition of the aforementioned transformed country characteristics. In lagged models, we only lagged the country's democracy score. For instance, a one-year lag would produce the model equation $\log(HHI_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + ... + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$, where $X_{1i(t-1)}$ is the country's democracy score in the prior year (year t-1).

Additional predictors in this model include the African country's GNI, population, total ODAG, total imports and exports all on the log scale. Quadratic terms were included for the log-transformed GNI, population, and total ODAG. The log-transformed population variable added a cubic term to capture further nonlinear tendencies. In addition to these variables, colonial past, region, a cold war indicator and an interaction between cold war and ideology (i.e., pre-cold war ideology) were included.

$$Y_{it} \sim \mathrm{N}(\mu_{it}, \sigma^2)$$

$$\log(HHI_{it}) = \beta_0 + \beta_1 X_{1ik} + ... + \gamma_{0i} + \gamma_{tk} + \epsilon_{it}$$

$$\gamma_{0i} \sim \mathrm{N}(0, \sigma^2_{country})$$

$$\gamma_{0t} \sim \mathrm{N}(0, \sigma^2_{year})$$

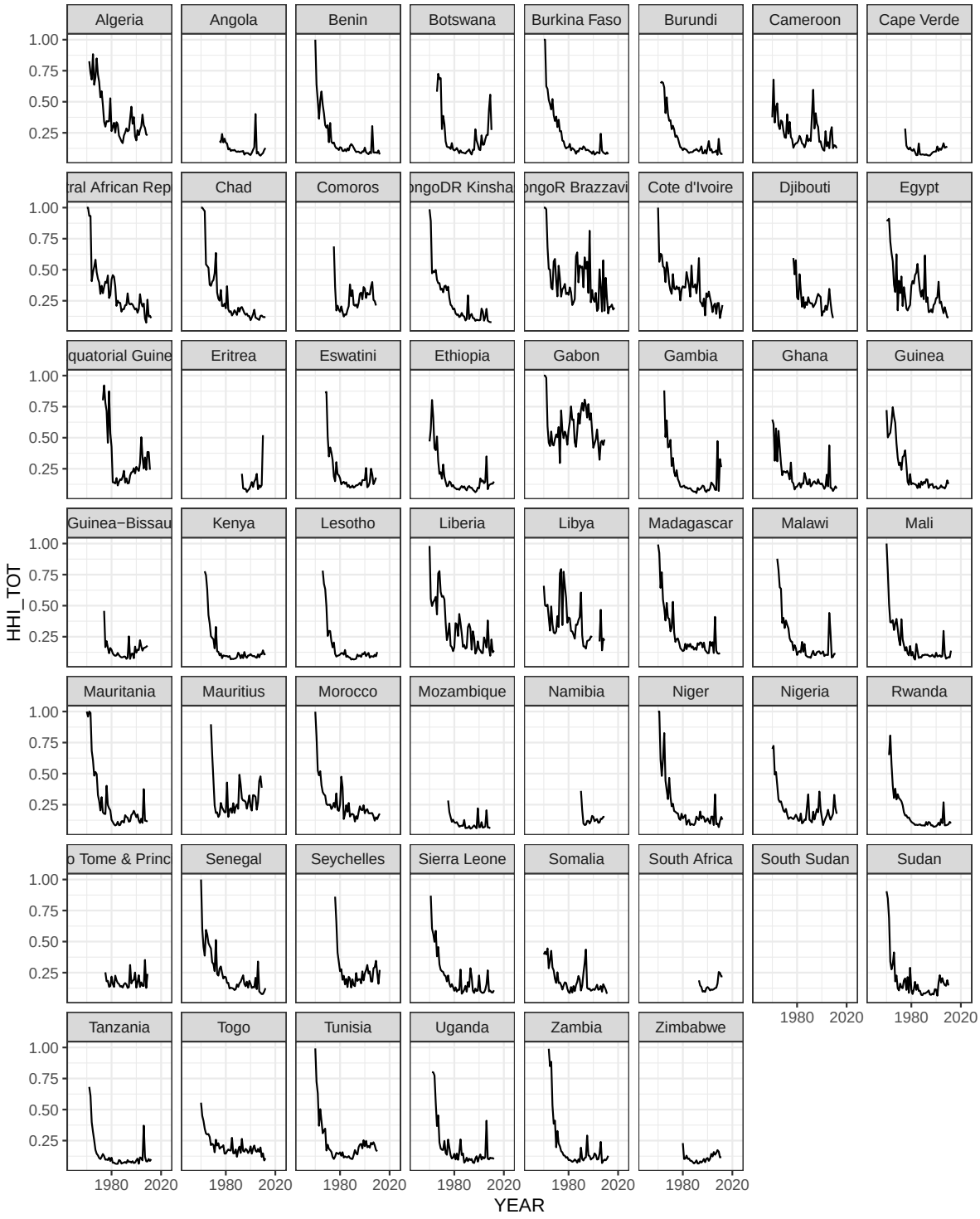
Data Visualizations and Results

```
[1] NaN
```

```
[1] 0.2679758
```

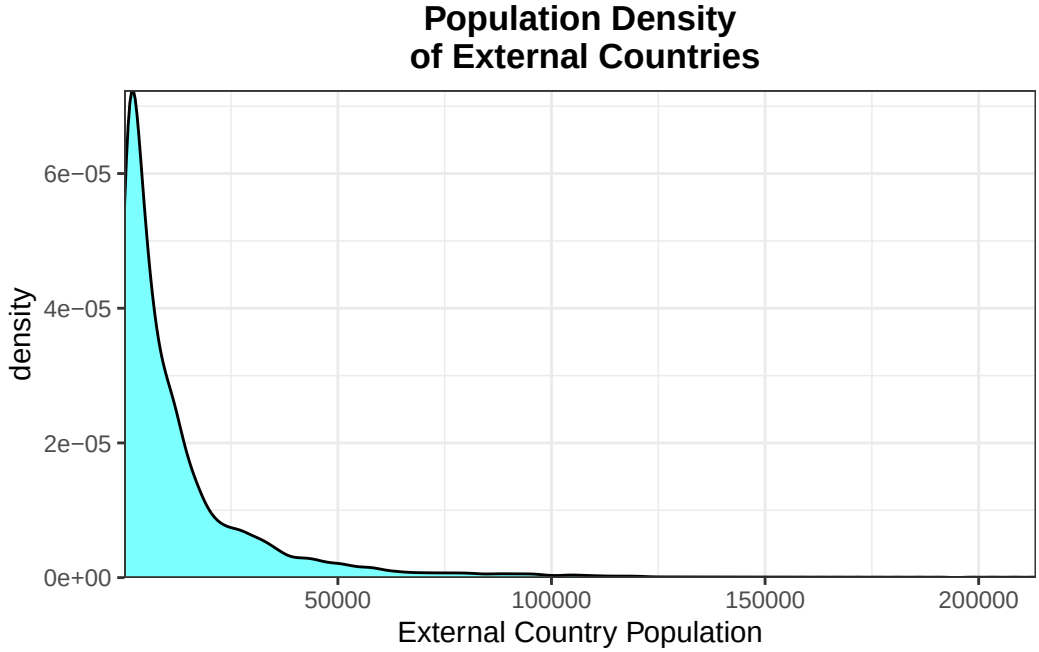
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[1] 0.07734139
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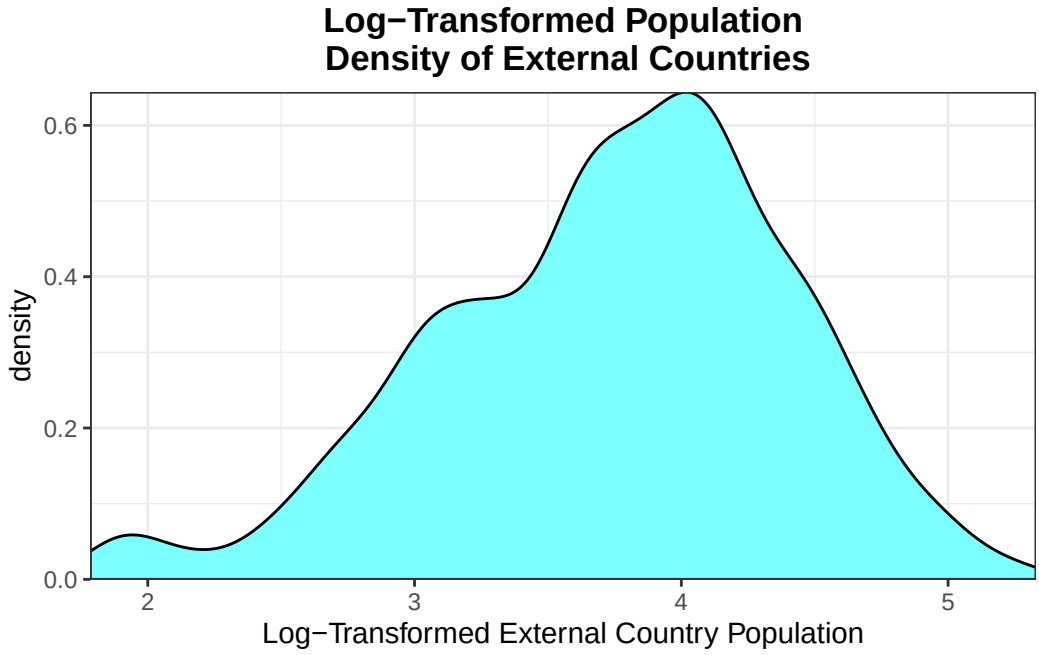
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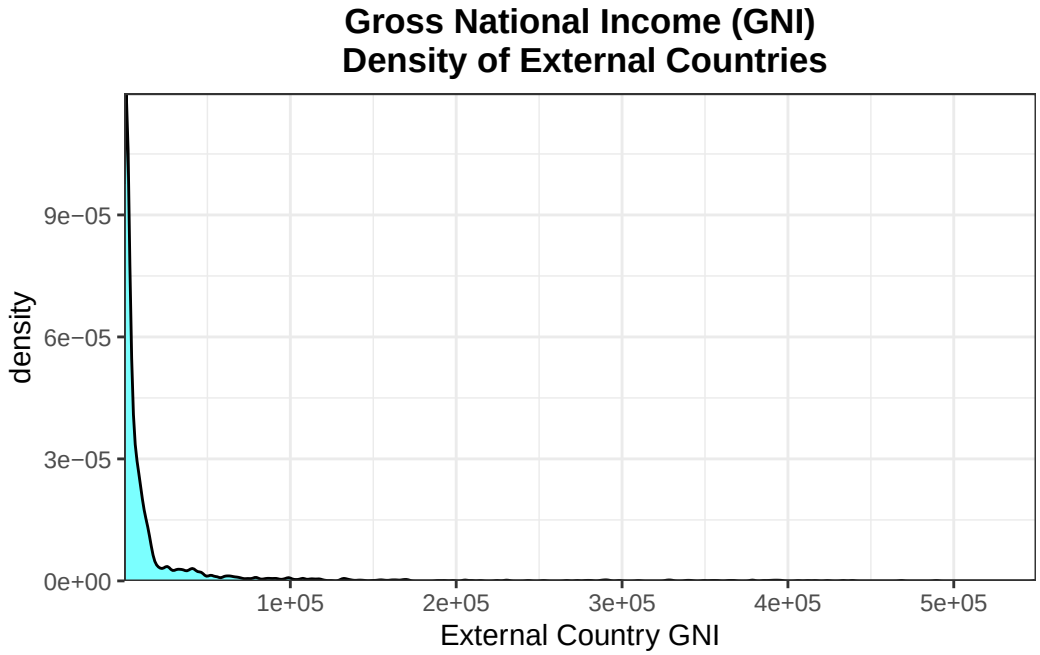


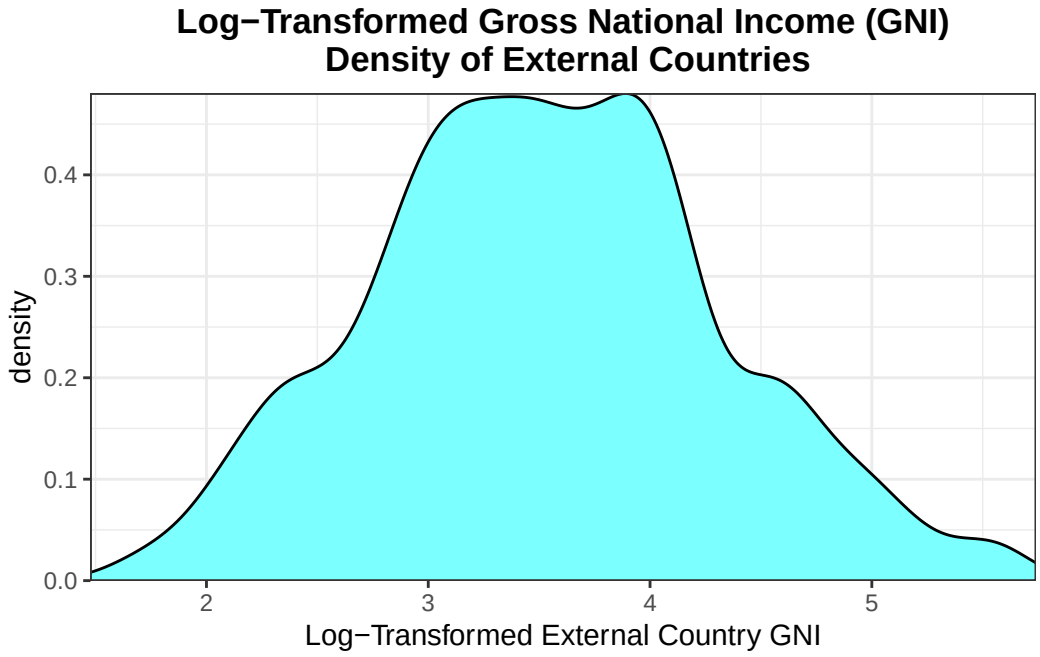
The distributions of the full model predictor variables and their log transformations are plotted

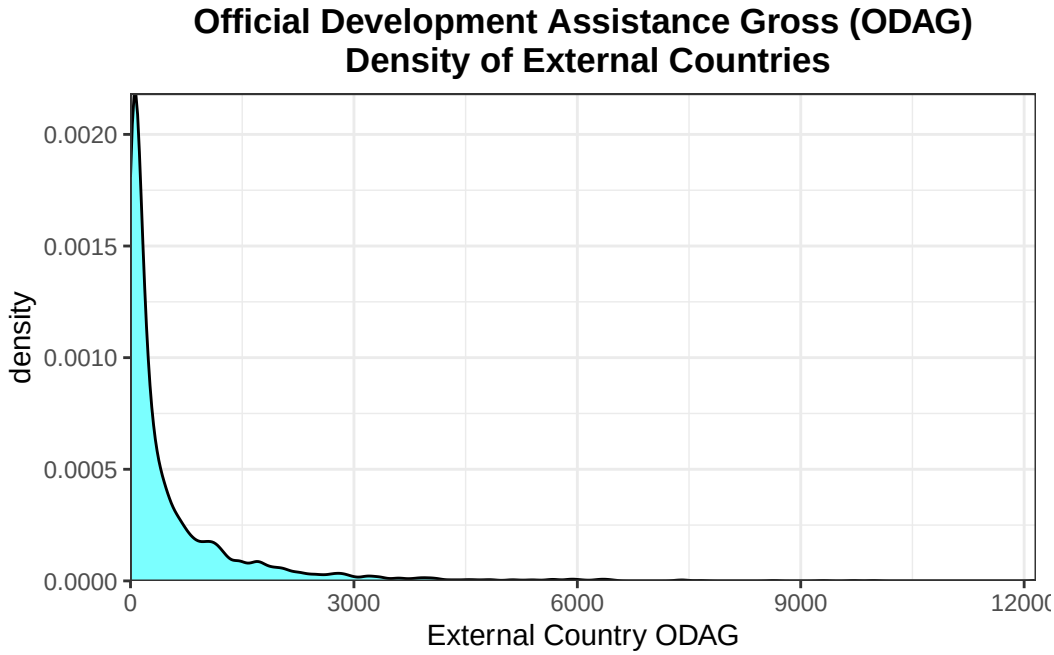
below. The log-transformed variables all display a noticeably reduced right-skew, improving their performance and interpretation in the full models.

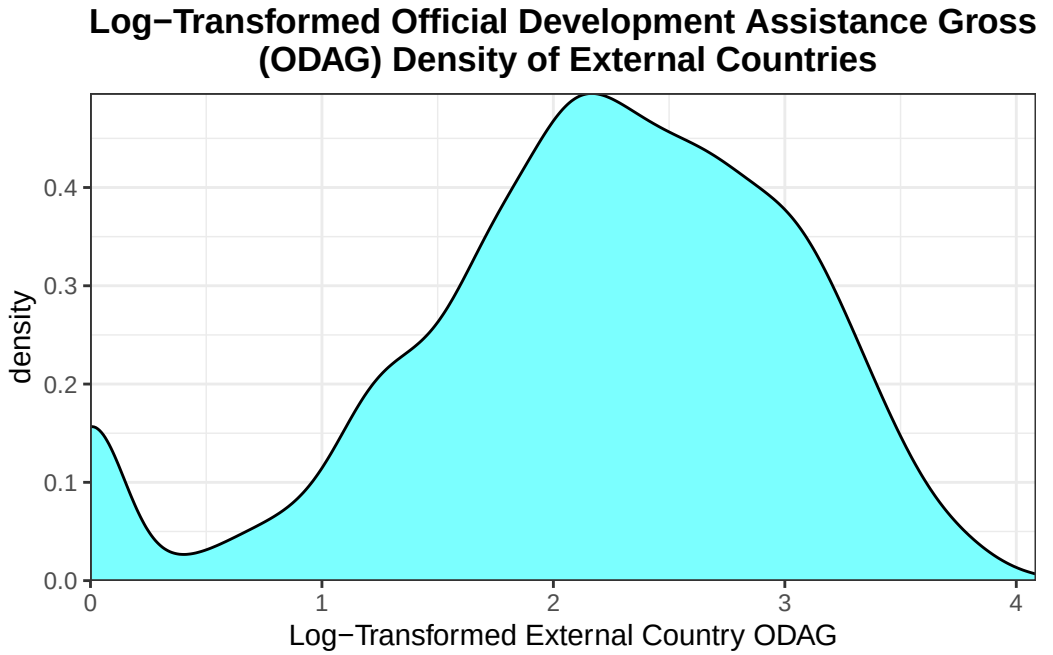





Gross National Income (GNI)
Density of External Countries
density
9e−05
6e−05
3e−05
0e+00
1e+05
2e+05
3e+05
4e+05
5e+05
External Country GNI


Log−Transformed Gross National Income (GNI)
Density of External Countries
density
0.4
0.3
0.2
0.1
0.0
2
3
4
5
Log−Transformed External Country GNI





The relationship between relevant democracy scores and log-transformed HHI, grouped and colored by year, and the relationship between log-transformed full model predictor variables and HHI are all visualized below.

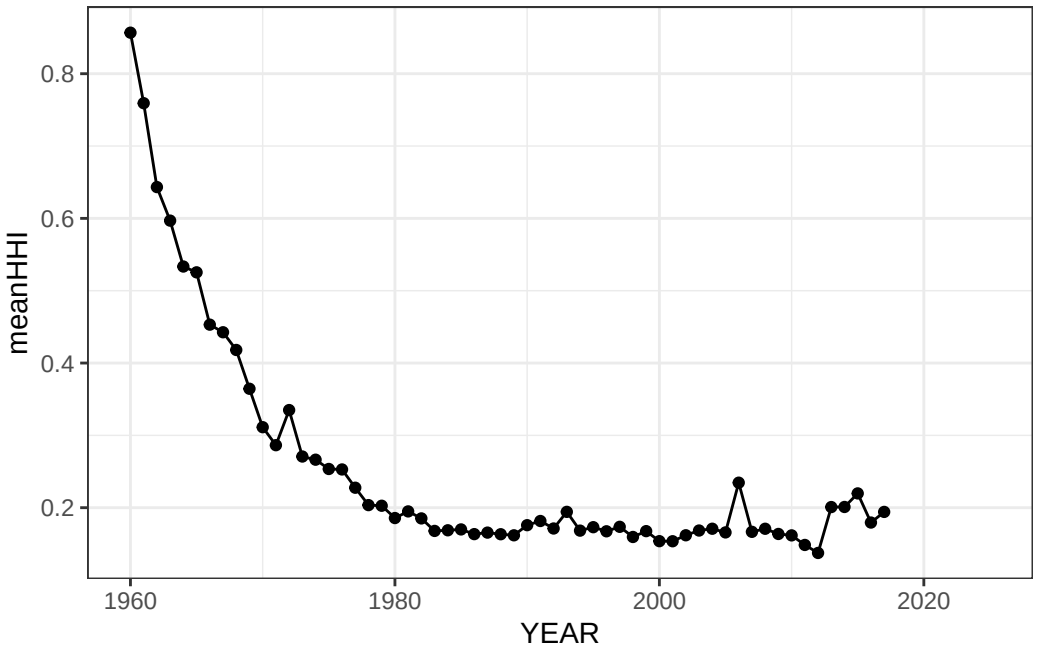
```
# A tibble: 54 x 2
   COUNTRY           TOTLIB1
   <chr>               <dbl>
 1 Somalia              13.8
 2 Equatorial Guinea    13.6
 3 South Sudan          13.3
 4 Libya                13.1
 5 Eritrea              13.0
 6 Sudan                12.9
 7 CongoDR Kinshasa     12.6
 8 Chad                 12.5
 9 Burundi              12.3
10 Angola               12.2
# i 44 more rows
```

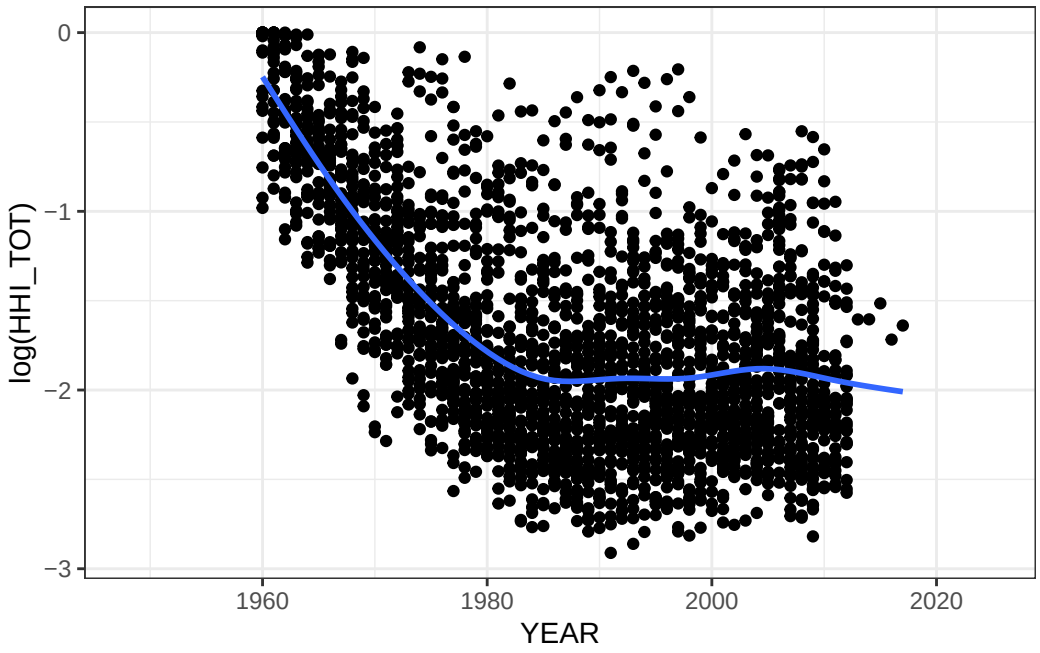
```
# A tibble: 54 x 2
   COUNTRY             V2X_LIBDEM
   <chr>                    <dbl>
 1 Mauritius                0.62
 2 Botswana                 0.536
 3 Namibia                  0.511
 4 Cape Verde               0.487
 5 Senegal                  0.453
 6 Sao Tome & Principe      0.350
 7 Ghana                    0.342
 8 South Africa             0.341
 9 Benin                    0.338
10 Seychelles               0.295
# i 44 more rows
```

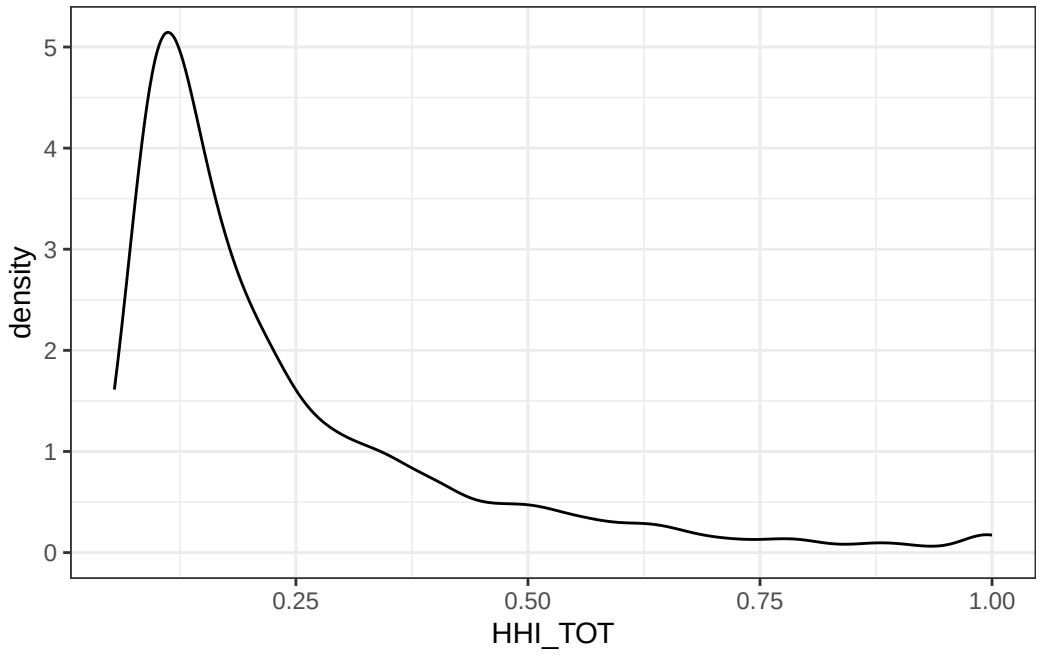
```
# A tibble: 52 x 2
   COUNTRY      POLITY2
   <chr>          <dbl>
 1 Mauritius      9.73
 2 Botswana       8.51
 3 Namibia        6.62
 4 South Africa   5.87
 5 Cape Verde     4.61
 6 Gambia         2.06
 7 Comoros        1.77
 8 Lesotho        1.08
 9 Madagascar     0.915
```

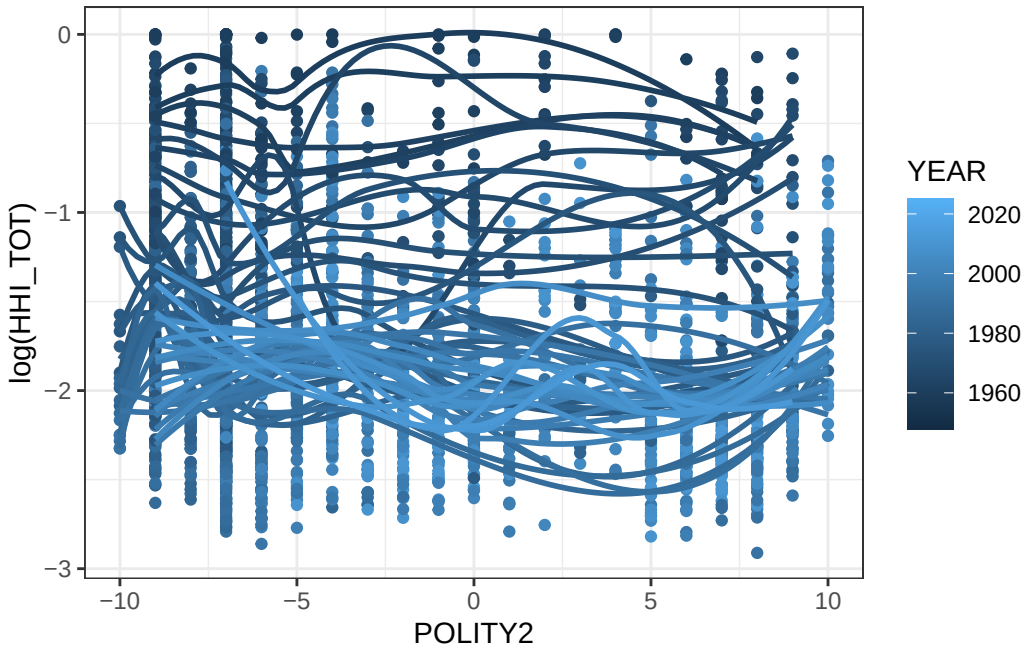
```
10 Benin          0.288
# i 42 more rows

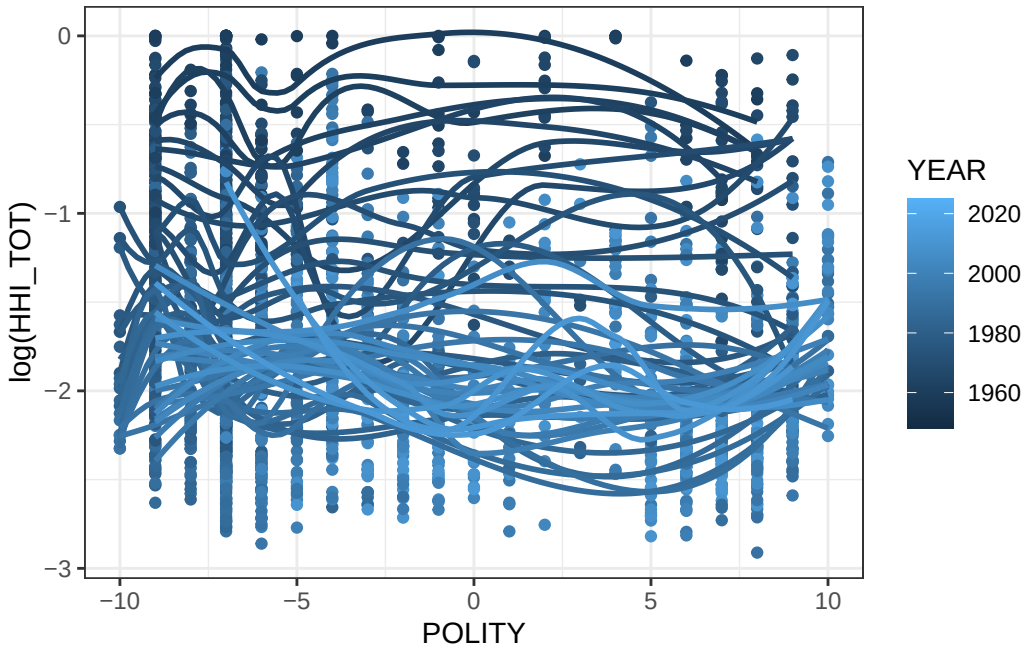

# A tibble: 66 x 2
    YEAR meanHHI
   <dbl>   <dbl>
 1  1960   0.857
 2  1961   0.759
 3  1962   0.643
 4  1963   0.597
 5  1964   0.533
 6  1965   0.525
 7  1966   0.453
 8  1967   0.443
 9  1968   0.418
10  1969   0.364
# i 56 more rows
```

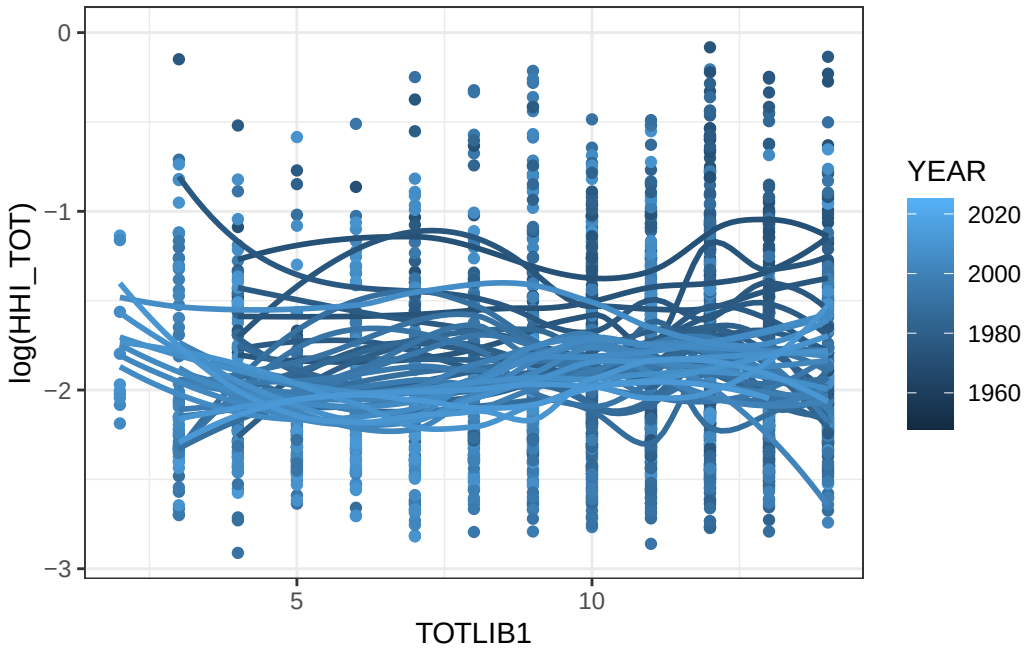


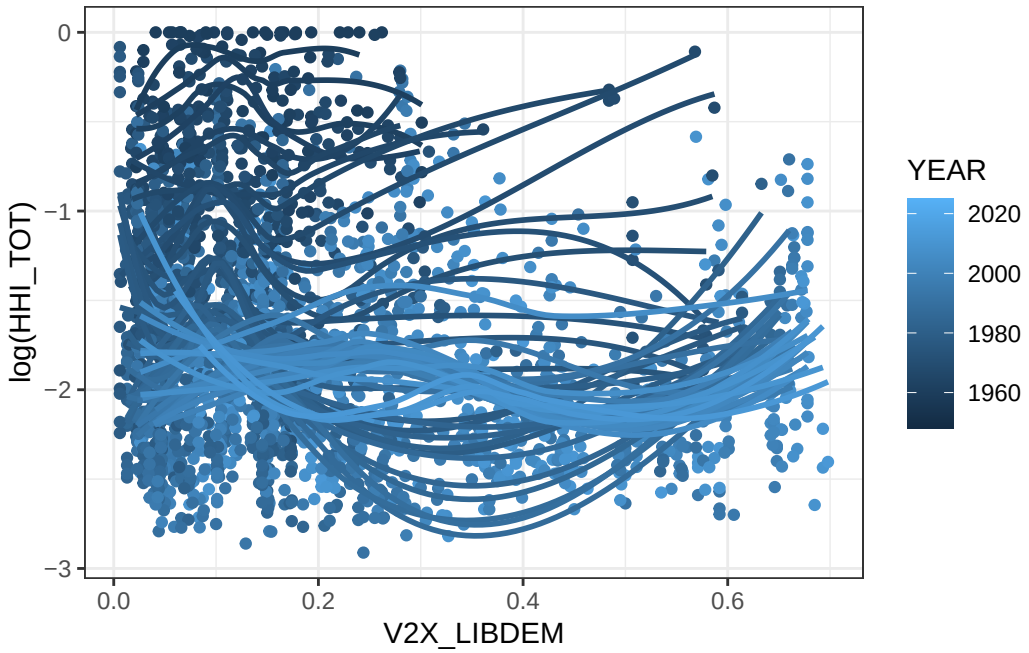

0
−1
−2
−3
log(HHI_TOT)
1960
1980
2000
2020
YEAR

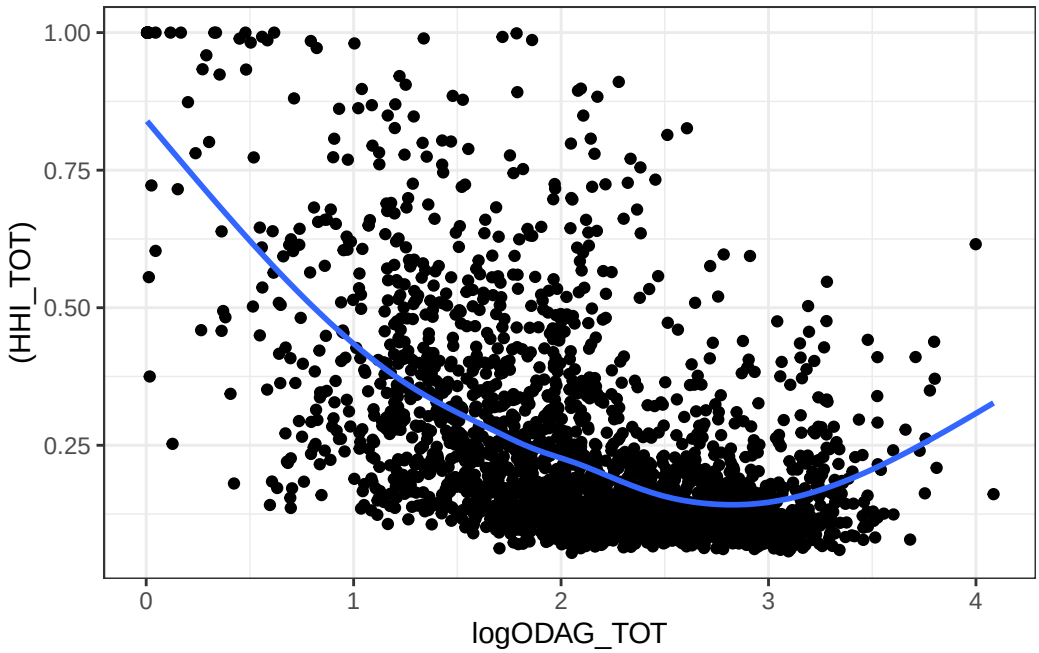

5
4
3
2
1
0
density
0.25
0.50
0.75
1.00
HHI_TOT

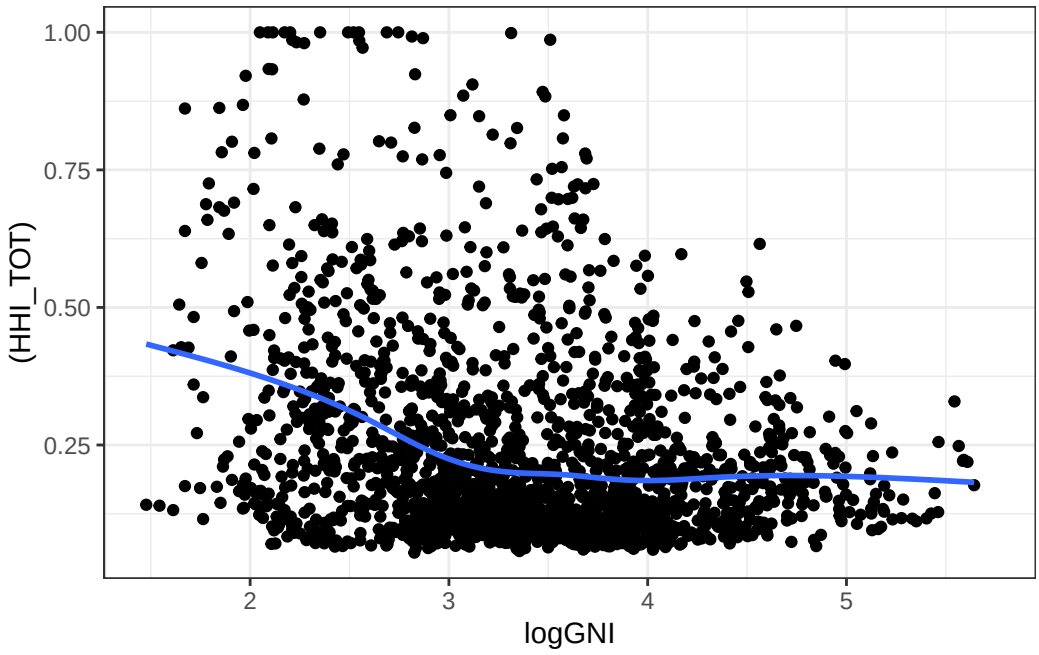

log(HHI_TOT)
0
−1
−2
−3
−10
−5
0
5
10
POLITY2
YEAR
2020
2000
1980
1960

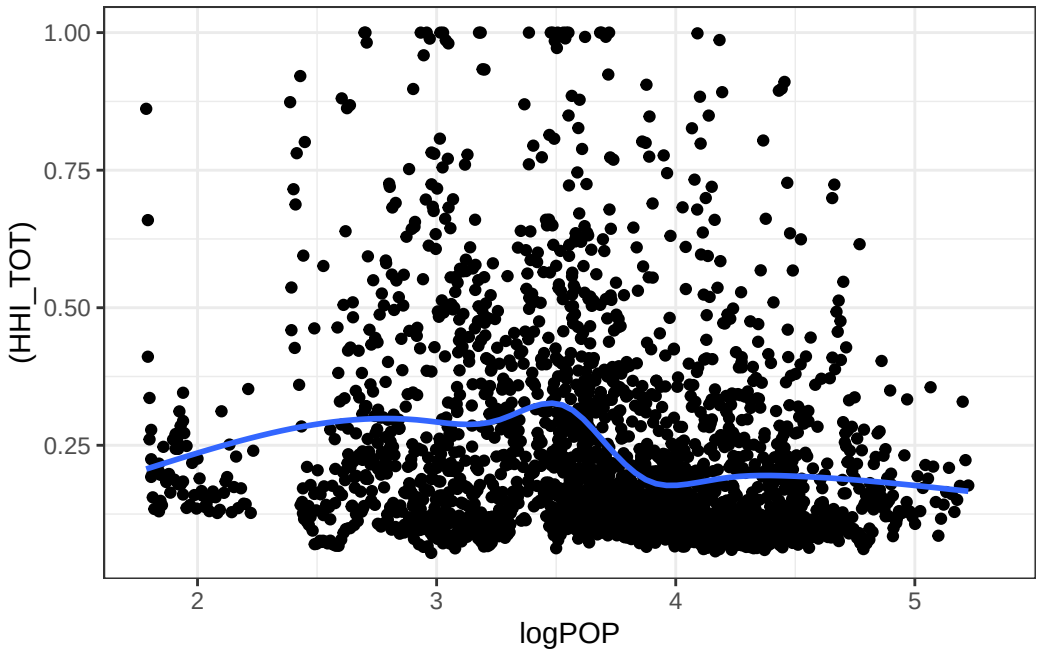

log(HHI_TOT)
0
−1
−2
−3
−10
−5
0
5
10
POLITY
YEAR
2020
2000
1980
1960


log(HHI_TOT)
0
−1
−2
−3
5
10
TOTLIB1
YEAR
2020
2000
1980
1960


log(HHI_TOT)
0
−1
−2
−3
0.0
0.2
0.4
0.6
V2X_LIBDEM
YEAR
2020
2000
1980
1960


1.00
0.75
0.50
0.25
(HHI_TOT)
0
1
2
3
4
logODAG_TOT


1.00
0.75
0.50
0.25
(HHI_TOT)
2
3
4
5
logGNI



Summaries for the reduced and full models for ODAG are shown with the regression coefficient estimates in the column labeled "Est." starred by significance as denoted in the legend below. Next to each coefficient estimate is their respective standard error.

HHI Countries Results

HHI IO Results

HHI Total Results

Table 1: **HHI Countries: Reduced Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.145***	0.076	−1.147***	0.076	−1.150***	0.075	−1.163***	0.075
V2X_LIBDEM	−0.280**	0.092						
V2X_LIBDEM_lag1			−0.322***	0.093				
V2X_LIBDEM_lag2					−0.351***	0.094		
V2X_LIBDEM_lag3							−0.327***	0.095
SD (Intercept YEAR)	0.377		0.365		0.353		0.344	
SD (Intercept CCODE)	0.391		0.396		0.400		0.405	
SD (Observations)	0.343		0.339		0.336		0.333	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 2: **HHI Countries: Reduced Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.220***	0.075	−1.234***	0.075	−1.247***	0.074	−1.255***	0.073
POLITY2	−0.002	0.002						
POLITY2_lag1			−0.004*	0.002				
POLITY2_lag2					−0.005**	0.002		
POLITY2_lag3							−0.006**	0.002
SD (Intercept YEAR)	0.378		0.368		0.350		0.340	
SD (Intercept CCODE)	0.391		0.396		0.399		0.404	
SD (Observations)	0.343		0.340		0.337		0.335	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 3: **HHI Countries: Reduced Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.457***	0.075	−1.474***	0.074	−1.478***	0.073	−1.501***	0.073
TOTLIB1	0.007+	0.004						
TOTLIB1_lag1			0.007+	0.004				
TOTLIB1_lag2					0.006	0.004		
TOTLIB1_lag3							0.007+	0.004
SD (Intercept YEAR)	0.168		0.132		0.113		0.098	
SD (Intercept CCODE)	0.425		0.430		0.433		0.435	
SD (Observations)	0.330		0.326		0.324		0.324	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 4: **HHI Countries: Full Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.
Intercept	−14.668***	1.902	−14.885***	1.900	−15.322***	1.909	−15.541***
V2X_LIBDEM	−0.172+	0.095					
V2X_LIBDEM_lag1			−0.232*	0.095			
V2X_LIBDEM_lag2					−0.279**	0.095	
V2X_LIBDEM_lag3							−0.271**
logPOP	12.840***	1.575	12.961***	1.575	13.307***	1.583	13.472***
logPOP^2	−3.122***	0.424	−3.194***	0.424	−3.321***	0.427	−3.387***
logPOP^3	0.226***	0.038	0.235***	0.038	0.249***	0.038	0.257***
logGNI	−1.278***	0.156	−1.097***	0.158	−0.917***	0.161	−0.792***
logGNI^2	0.195***	0.022	0.171***	0.022	0.148***	0.022	0.131***
logODAG_COUNTRIES	−0.393***	0.080	−0.456***	0.084	−0.574***	0.087	−0.660***
logODAG_COUNTRIES^2	0.118***	0.020	0.132***	0.020	0.156***	0.021	0.173***
logTRDIMtot	0.014	0.027	0.025	0.027	0.025	0.028	0.028
logTRDEXtot	0.025	0.026	0.022	0.026	0.023	0.026	0.018
COLD_WAR	0.182*	0.085	0.176*	0.084	0.161*	0.081	0.155*
CW_Socialist	−0.054	0.043	−0.065	0.042	−0.061	0.042	−0.065
CW_Marxist	0.098*	0.044	0.090*	0.044	0.081+	0.043	0.076+
COL_British	0.058	0.183	0.043	0.179	0.022	0.178	0.001
COL_Portuguese	−0.204	0.249	−0.217	0.244	−0.222	0.241	−0.233
COL_Spanish	−0.282	0.485	−0.300	0.475	−0.326	0.470	−0.341
COL_Italian	−0.864*	0.364	−0.869*	0.356	−0.855*	0.353	−0.836*
COL_Belgian	0.313	0.492	0.288	0.482	0.271	0.477	0.256
COL_German	−0.417+	0.224	−0.420+	0.219	−0.422+	0.217	−0.431*
COL_US	−0.103	0.464	−0.097	0.454	−0.097	0.450	−0.092
COL_None	0.262	0.498	0.235	0.488	0.206	0.483	0.179
REG_E_Africa	−0.057	0.268	−0.069	0.263	−0.071	0.261	−0.079
REG_S_Africa	−0.806**	0.285	−0.797**	0.279	−0.789**	0.277	−0.788**
REG_C_Africa	−0.043	0.290	−0.042	0.284	−0.045	0.281	−0.046
REG_W_Africa	−0.478+	0.250	−0.473+	0.245	−0.467+	0.243	−0.473+
SD (Intercept YEAR)	0.268		0.262		0.252		0.241
SD (Intercept CCODE)	0.439		0.430		0.425		0.424
SD (Observations)	0.318		0.314		0.311		0.309

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 5: **HHI Countries: Full Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.
Intercept	−22.705***	2.338	−22.626***	2.361	−22.971***	2.391	−22.807***
POLITY2	0.004*	0.002					
POLITY2_lag1			0.002	0.002			
POLITY2_lag2					0.000	0.002	
POLITY2_lag3							−0.001
logPOP	18.685***	1.891	18.587***	1.910	18.856***	1.933	18.712***
logPOP^2	−4.569***	0.494	−4.569***	0.499	−4.665***	0.504	−4.645***
logPOP^3	0.349***	0.043	0.351***	0.044	0.361***	0.044	0.361***
logGNI	−1.347***	0.164	−1.215***	0.167	−1.066***	0.169	−0.947***
logGNI^2	0.203***	0.022	0.184***	0.023	0.164***	0.023	0.148***
logODAG_COUNTRIES	−0.381***	0.082	−0.443***	0.086	−0.559***	0.091	−0.653***
logODAG_COUNTRIES^2	0.114***	0.020	0.127***	0.021	0.151***	0.021	0.169***
logTRDIMtot	0.025	0.028	0.031	0.028	0.027	0.028	0.028
logTRDEXtot	0.016	0.026	0.016	0.026	0.018	0.027	0.016
COLD_WAR	0.296***	0.087	0.275**	0.085	0.254**	0.082	0.241**
CW_Socialist	−0.041	0.043	−0.060	0.043	−0.061	0.043	−0.062
CW_Marxist	0.131**	0.044	0.118**	0.044	0.106*	0.044	0.097*
COL_British	−0.136	0.143	−0.139	0.141	−0.153	0.140	−0.169
COL_Portuguese	−0.364+	0.207	−0.367+	0.205	−0.367+	0.204	−0.376+
COL_Spanish	0.163	0.379	0.140	0.375	0.119	0.372	0.095
COL_Italian	−0.679*	0.283	−0.700*	0.280	−0.693*	0.278	−0.683*
COL_Belgian	0.195	0.379	0.185	0.375	0.177	0.372	0.160
COL_German	−0.452**	0.171	−0.455**	0.169	−0.457**	0.168	−0.465**
COL_US	−0.123	0.354	−0.117	0.350	−0.116	0.347	−0.111
COL_None	0.121	0.384	0.109	0.380	0.087	0.377	0.060
REG_E_Africa	−0.074	0.208	−0.080	0.205	−0.076	0.204	−0.075
REG_S_Africa	−0.572*	0.223	−0.572**	0.221	−0.568**	0.220	−0.562*
REG_C_Africa	−0.001	0.227	−0.006	0.225	−0.012	0.223	−0.009
REG_W_Africa	−0.335+	0.195	−0.334+	0.193	−0.330+	0.191	−0.328+
SD (Intercept YEAR)	0.280		0.272		0.260		0.252
SD (Intercept CCODE)	0.333		0.329		0.326		0.327
SD (Observations)	0.317		0.315		0.312		0.309

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 6: **HHI Countries: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.
Intercept	−14.124***	1.973	−14.728***	2.003	−15.245***	2.052	−16.479***
TOTLIB1	0.007	0.004					
TOTLIB1_lag1			0.009*	0.004			
TOTLIB1_lag2					0.009*	0.004	
TOTLIB1_lag3							0.011*
logPOP	12.220***	1.677	12.647***	1.701	12.920***	1.742	13.838***
logPOP^2	−3.141***	0.455	−3.302***	0.462	−3.417***	0.474	−3.667***
logPOP^3	0.245***	0.041	0.262***	0.042	0.275***	0.043	0.296***
logGNI	−0.658***	0.175	−0.497**	0.179	−0.348+	0.184	−0.262
logGNI^2	0.116***	0.024	0.094***	0.025	0.075**	0.025	0.064*
logODAG_COUNTRIES	−0.772***	0.118	−0.742***	0.123	−0.637***	0.133	−0.634***
logODAG_COUNTRIES^2	0.196***	0.026	0.192***	0.027	0.171***	0.029	0.171***
logTRDIMtot	0.079**	0.029	0.085**	0.029	0.082**	0.029	0.069*
logTRDEXtot	−0.021	0.028	−0.017	0.028	−0.012	0.028	0.003
COLD_WAR	0.049	0.053	0.017	0.044	0.003	0.041	−0.016
CW_Socialist	−0.119*	0.052	−0.131*	0.052	−0.136**	0.052	−0.141**
CW_Marxist	0.066	0.046	0.068	0.046	0.074	0.047	0.075
COL_British	−0.070	0.166	−0.069	0.166	−0.071	0.166	−0.063
COL_Portuguese	−0.290	0.225	−0.301	0.225	−0.303	0.225	−0.277
COL_Spanish	−0.375	0.438	−0.393	0.438	−0.387	0.438	−0.415
COL_Italian	−0.708*	0.329	−0.718*	0.329	−0.688*	0.329	−0.701*
COL_Belgian	0.097	0.446	0.105	0.446	0.085	0.446	0.088
COL_German	−0.450*	0.202	−0.445*	0.202	−0.445*	0.202	−0.447*
COL_US	−0.046	0.419	−0.057	0.419	−0.057	0.419	−0.067
COL_None	0.078	0.451	0.088	0.451	0.072	0.451	0.071
REG_E_Africa	−0.031	0.243	−0.060	0.243	−0.068	0.243	−0.059
REG_S_Africa	−0.648*	0.259	−0.665*	0.258	−0.674**	0.259	−0.701**
REG_C_Africa	0.097	0.263	0.072	0.262	0.063	0.262	0.047
REG_W_Africa	−0.364	0.227	−0.380+	0.227	−0.385+	0.227	−0.399+
SD (Intercept YEAR)	0.140		0.104		0.094		0.081
SD (Intercept CCODE)	0.395		0.395		0.395		0.407
SD (Observations)	0.306		0.303		0.302		0.301

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 7: **HHI IO: Reduced Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.030***	0.068	−1.033***	0.066	−1.030***	0.066	−1.038***	0.065
V2X_LIBDEM	−0.121+	0.071						
V2X_LIBDEM_lag1			−0.211**	0.072				
V2X_LIBDEM_lag2					−0.256***	0.073		
V2X_LIBDEM_lag3							−0.256***	0.074
SD (Intercept YEAR)	0.452		0.429		0.417		0.405	
SD (Intercept CCODE)	0.254		0.261		0.264		0.266	
SD (Observations)	0.337		0.335		0.335		0.337	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 8: **HHI IO: Reduced Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.034***	0.071	−1.069***	0.068	−1.086***	0.066	−1.101***	0.065
POLITY2	0.000	0.002						
POLITY2_lag1			−0.002	0.002				
POLITY2_lag2					−0.004*	0.002		
POLITY2_lag3							−0.005**	0.002
SD (Intercept YEAR)	0.472		0.447		0.434		0.419	
SD (Intercept CCODE)	0.243		0.243		0.245		0.249	
SD (Observations)	0.334		0.334		0.335		0.336	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 9: **HHI IO: Reduced Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.265***	0.053	−1.303***	0.053	−1.323***	0.053	−1.323***	0.053
TOTLIB1	0.001	0.003						
TOTLIB1_lag1			0.004	0.003				
TOTLIB1_lag2					0.005	0.003		
TOTLIB1_lag3							0.004	0.003
SD (Intercept YEAR)	0.132		0.123		0.111		0.103	
SD (Intercept CCODE)	0.264		0.270		0.274		0.279	
SD (Observations)	0.344		0.341		0.339		0.337	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 10: **HHI IO: Full Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−1.800	1.799	−1.542	1.801	−1.099	1.823	−0.966	1.851
V2X_LIBDEM	−0.300***	0.079						
V2X_LIBDEM_lag1			−0.398***	0.079				
V2X_LIBDEM_lag2					−0.449***	0.080		
V2X_LIBDEM_lag3							−0.425***	0.080
logPOP	5.402***	1.351	5.128***	1.356	4.849***	1.371	4.779***	1.393
logPOP^2	−1.523***	0.349	−1.467***	0.351	−1.410***	0.355	−1.403***	0.360
logPOP^3	0.098**	0.031	0.095**	0.031	0.091**	0.031	0.091**	0.032
logGNI	−1.268***	0.135	−1.224***	0.137	−1.203***	0.139	−1.190***	0.142
logGNI^2	0.204***	0.018	0.198***	0.018	0.195***	0.018	0.193***	0.019
logTRDIMtot	−0.051*	0.025	−0.046+	0.025	−0.039	0.025	−0.034	0.026
logTRDEXtot	0.013	0.023	0.019	0.024	0.017	0.024	0.014	0.025
COLD_WAR	−0.106	0.085	−0.100	0.088	−0.118	0.088	−0.125	0.087
CW_Socialist	−0.063	0.042	−0.070+	0.042	−0.068	0.042	−0.064	0.042
CW_Marxist	−0.108**	0.041	−0.108**	0.041	−0.110**	0.041	−0.108**	0.041
COL_British	0.339	0.372	0.337	0.369	0.343	0.378	0.349	0.383
COL_Portuguese	−0.504	0.515	−0.512	0.511	−0.532	0.523	−0.543	0.530
COL_Spanish	−1.093	1.033	−1.089	1.025	−1.118	1.049	−1.128	1.064
COL_Italian	−0.688	0.750	−0.702	0.744	−0.736	0.762	−0.747	0.773
COL_Belgian	2.046*	1.039	2.015+	1.031	2.045+	1.055	2.066+	1.070
COL_German	0.487	0.476	0.487	0.472	0.496	0.483	0.501	0.490
COL_US	−0.486	0.995	−0.478	0.987	−0.487	1.010	−0.493	1.025
COL_None	1.942+	1.035	1.928+	1.026	1.957+	1.050	1.972+	1.065
COL_Unique	1.087	1.029	1.099	1.020	1.111	1.044	1.115	1.059
REG_E_Africa	−1.351*	0.544	−1.360*	0.540	−1.382*	0.552	−1.389*	0.560
REG_S_Africa	−1.298*	0.594	−1.279*	0.589	−1.295*	0.603	−1.307*	0.612
REG_C_Africa	−1.744**	0.601	−1.735**	0.596	−1.763**	0.610	−1.777**	0.619
REG_W_Africa	−1.281*	0.523	−1.268*	0.518	−1.281*	0.531	−1.291*	0.538
SD (Intercept YEAR)	0.256		0.270		0.269		0.264	
SD (Intercept CCODE)	0.950		0.942		0.964		0.978	
SD (Observations)	0.313		0.311		0.311		0.312	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 11: **HHI IO: Full Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−0.789	2.117	−1.118	2.079	−0.801	2.045	0.074	2.016
POLITY2	−0.003	0.002						
POLITY2_lag1			−0.005**	0.002				
POLITY2_lag2					−0.007***	0.002		
POLITY2_lag3							−0.007***	0.002
logPOP	5.439***	1.585	5.242***	1.566	5.117***	1.536	4.358**	1.514
logPOP^2	−1.632***	0.410	−1.549***	0.405	−1.537***	0.397	−1.351***	0.391
logPOP^3	0.111**	0.036	0.104**	0.036	0.103**	0.035	0.088*	0.034
logGNI	−1.264***	0.147	−1.210***	0.147	−1.200***	0.147	−1.154***	0.147
logGNI^2	0.199***	0.020	0.193***	0.019	0.191***	0.019	0.187***	0.019
logTRDIMtot	−0.041	0.026	−0.040	0.026	−0.036	0.026	−0.038	0.027
logTRDEXtot	0.003	0.025	0.008	0.025	0.006	0.025	0.010	0.025
COLD_WAR	−0.249**	0.079	−0.191*	0.083	−0.214**	0.082	−0.208*	0.082
CW_Socialist	−0.071+	0.043	−0.081+	0.043	−0.089*	0.043	−0.082+	0.043
CW_Marxist	−0.099*	0.042	−0.102*	0.042	−0.112**	0.042	−0.117**	0.042
COL_British	0.551	0.440	0.527	0.408	0.552	0.419	0.555	0.417
COL_Portuguese	−0.145	0.662	−0.123	0.614	−0.118	0.631	−0.114	0.627
COL_Spanish	−1.577	1.203	−1.421	1.117	−1.463	1.147	−1.470	1.140
COL_Italian	−0.829	0.866	−0.780	0.804	−0.815	0.826	−0.797	0.821
COL_Belgian	2.191+	1.205	2.041+	1.119	2.083+	1.149	2.060+	1.142
COL_German	0.499	0.550	0.470	0.511	0.481	0.525	0.475	0.522
COL_US	−0.481	1.149	−0.442	1.066	−0.449	1.095	−0.446	1.089
COL_None	2.170+	1.194	2.052+	1.109	2.113+	1.138	2.104+	1.132
COL_Unique	1.327	1.190	1.204	1.104	1.093	1.133	1.074	1.127
REG_E_Africa	−1.419*	0.630	−1.358*	0.585	−1.395*	0.600	−1.381*	0.597
REG_S_Africa	−1.633*	0.694	−1.548*	0.644	−1.601*	0.661	−1.593*	0.658
REG_C_Africa	−1.710*	0.708	−1.619*	0.658	−1.662*	0.675	−1.651*	0.671
REG_W_Africa	−1.515*	0.605	−1.431*	0.562	−1.470*	0.577	−1.454*	0.573
SD (Intercept YEAR)	0.211		0.234		0.231		0.233	
SD (Intercept CCODE)	1.096		1.016		1.044		1.038	
SD (Observations)	0.312		0.313		0.311		0.312	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 12: **HHI IO: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−1.534	1.931	−1.056	1.954	−0.736	1.988	−0.934	2.002
TOTLIB1	0.005	0.004						
TOTLIB1_lag1			0.008*	0.004				
TOTLIB1_lag2					0.010**	0.004		
TOTLIB1_lag3							0.010*	0.004
logPOP	4.623**	1.505	4.035**	1.529	3.606*	1.559	3.376*	1.583
logPOP^2	−1.414***	0.390	−1.298**	0.396	−1.195**	0.404	−1.114**	0.411
logPOP^3	0.101**	0.034	0.094**	0.035	0.086*	0.036	0.080*	0.036
logGNI	−1.041***	0.155	−0.913***	0.159	−0.807***	0.162	−0.720***	0.165
logGNI^2	0.173***	0.020	0.156***	0.021	0.142***	0.021	0.129***	0.022
logTRDIMtot	−0.052+	0.027	−0.048+	0.027	−0.042	0.027	−0.051+	0.027
logTRDEXtot	0.014	0.025	0.019	0.026	0.019	0.026	0.029	0.026
COLD_WAR	−0.217***	0.053	−0.213***	0.053	−0.228***	0.051	−0.218***	0.049
CW_Socialist	−0.047	0.051	−0.052	0.051	−0.042	0.051	−0.025	0.052
CW_Marxist	−0.088+	0.045	−0.089+	0.045	−0.090*	0.046	−0.076+	0.046
COL_British	0.273	0.343	0.262	0.340	0.270	0.340	0.272	0.317
COL_Portuguese	−0.526	0.475	−0.544	0.470	−0.540	0.471	−0.485	0.439
COL_Spanish	−0.934	0.951	−0.909	0.942	−0.913	0.943	−0.836	0.877
COL_Italian	−0.532	0.691	−0.525	0.684	−0.537	0.686	−0.494	0.638
COL_Belgian	1.807+	0.958	1.755+	0.949	1.745+	0.950	1.619+	0.884
COL_German	0.441	0.438	0.434	0.434	0.431	0.435	0.403	0.404
COL_US	−0.433	0.916	−0.426	0.907	−0.420	0.908	−0.378	0.844
COL_None	1.717+	0.954	1.689+	0.945	1.680+	0.946	1.553+	0.881
COL_Unique	1.034	0.948	1.031	0.939	1.015	0.940	0.970	0.875
REG_E_Africa	−1.274*	0.501	−1.297**	0.497	−1.307**	0.497	−1.251**	0.463
REG_S_Africa	−1.193*	0.547	−1.177*	0.542	−1.186*	0.543	−1.162*	0.505
REG_C_Africa	−1.628**	0.554	−1.625**	0.549	−1.633**	0.549	−1.565**	0.511
REG_W_Africa	−1.217*	0.482	−1.214*	0.477	−1.223*	0.478	−1.198**	0.444
SD (Intercept YEAR)	0.120		0.122		0.114		0.106	
SD (Intercept CCODE)	0.874		0.865		0.866		0.805	
SD (Observations)	0.319		0.316		0.315		0.314	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 13: **HHI Total: Reduced Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.596***	0.083	−1.603***	0.081	−1.603***	0.080	−1.615***	0.079
V2X_LIBDEM	−0.209*	0.093						
V2X_LIBDEM_lag1			−0.236*	0.093				
V2X_LIBDEM_lag2					−0.289**	0.094		
V2X_LIBDEM_lag3							−0.275**	0.095
SD (Intercept YEAR)	0.495		0.479		0.464		0.453	
SD (Intercept CCODE)	0.334		0.340		0.344		0.349	
SD (Observations)	0.349		0.344		0.339		0.336	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 14: **HHI Total: Reduced Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.652***	0.082	−1.668***	0.080	−1.682***	0.079	−1.693***	0.078
POLITY2	−0.001	0.002						
POLITY2_lag1			−0.002	0.002				
POLITY2_lag2					−0.004+	0.002		
POLITY2_lag3							−0.005**	0.002
SD (Intercept YEAR)	0.495		0.481		0.461		0.449	
SD (Intercept CCODE)	0.336		0.341		0.344		0.348	
SD (Observations)	0.350		0.345		0.342		0.338	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 15: **HHI Total: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.866***	0.071	−1.897***	0.069	−1.914***	0.069	−1.921***	0.069
TOTLIB1	0.001	0.004						
TOTLIB1_lag1			0.002	0.004				
TOTLIB1_lag2					0.003	0.004		
TOTLIB1_lag3							0.003	0.004
SD (Intercept YEAR)	0.182		0.142		0.123		0.107	
SD (Intercept CCODE)	0.382		0.387		0.390		0.391	
SD (Observations)	0.330		0.325		0.323		0.322	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 16: **HHI Total: Full Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−8.299***	1.805	−9.542***	1.821	−10.825***	1.843	−11.765***	1.866
V2X_LIBDEM	−0.325***	0.097						
V2X_LIBDEM_lag1			−0.393***	0.097				
V2X_LIBDEM_lag2					−0.466***	0.097		
V2X_LIBDEM_lag3							−0.450***	0.097
logPOP	6.640***	1.517	7.653***	1.527	8.750***	1.541	9.486***	1.559
logPOP^2	−1.550***	0.413	−1.835***	0.414	−2.152***	0.418	−2.379***	0.423
logPOP^3	0.098**	0.037	0.123**	0.037	0.152***	0.038	0.175***	0.038
logGNI	−0.561***	0.156	−0.367*	0.157	−0.164	0.159	−0.022	0.160
logGNI^2	0.108***	0.022	0.081***	0.022	0.052*	0.022	0.033	0.022
logTRDIMtot	−0.017	0.028	0.001	0.028	0.015	0.028	0.019	0.028
logTRDEXtot	0.022	0.026	0.012	0.026	0.003	0.026	−0.003	0.027
COLD_WAR	0.437***	0.110	0.410***	0.107	0.379***	0.103	0.376***	0.101
CW_Socialist	−0.059	0.044	−0.065	0.043	−0.063	0.043	−0.064	0.043
CW_Marxist	0.013	0.045	0.006	0.044	−0.007	0.044	−0.010	0.043
COL_British	0.113	0.152	0.119	0.157	0.111	0.162	0.096	0.163
COL_Portuguese	−0.161	0.207	−0.145	0.213	−0.133	0.220	−0.133	0.222
COL_Spanish	−0.184	0.400	−0.215	0.414	−0.258	0.427	−0.263	0.431
COL_Italian	−0.588+	0.304	−0.626*	0.314	−0.647*	0.323	−0.636+	0.326
COL_Belgian	0.245	0.408	0.249	0.421	0.244	0.434	0.236	0.439
COL_German	−0.284	0.185	−0.297	0.191	−0.309	0.197	−0.313	0.199
COL_US	0.131	0.382	0.128	0.395	0.120	0.408	0.120	0.412
COL_None	0.236	0.414	0.230	0.427	0.206	0.440	0.180	0.444
REG_E_Africa	−0.295	0.223	−0.326	0.231	−0.349	0.237	−0.360	0.240
REG_S_Africa	−0.764**	0.238	−0.803**	0.246	−0.829**	0.253	−0.845***	0.255
REG_C_Africa	−0.217	0.242	−0.237	0.250	−0.257	0.257	−0.267	0.260
REG_W_Africa	−0.592**	0.209	−0.622**	0.216	−0.647**	0.222	−0.661**	0.224
SD (Intercept YEAR)	0.369		0.357		0.341		0.333	
SD (Intercept CCODE)	0.360		0.373		0.385		0.389	
SD (Observations)	0.327		0.321		0.314		0.311	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 17: **HHI Total: Full Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E
Intercept	−16.107***	2.334	−16.977***	2.344	−18.222***	2.367	−18.377***	2.39
POLITY2	0.003	0.002						
POLITY2_lag1			0.000	0.002				
POLITY2_lag2					−0.002	0.002		
POLITY2_lag3							−0.005*	0.00
logPOP	12.526***	1.895	13.217***	1.902	14.270***	1.918	14.359***	1.93
logPOP^2	−3.049***	0.496	−3.232***	0.497	−3.513***	0.501	−3.561***	0.50
logPOP^3	0.226***	0.044	0.242***	0.044	0.267***	0.044	0.273***	0.04
logGNI	−0.557***	0.165	−0.390*	0.166	−0.225	0.167	−0.082	0.16
logGNI^2	0.106***	0.023	0.082***	0.023	0.057*	0.023	0.038	0.02
logTRDIMtot	−0.012	0.028	0.002	0.028	0.013	0.029	0.013	0.02
logTRDEXtot	0.016	0.027	0.007	0.027	0.000	0.027	−0.002	0.02
COLD_WAR	0.518***	0.109	0.482***	0.106	0.441***	0.102	0.430***	0.10
CW_Socialist	−0.046	0.044	−0.061	0.044	−0.061	0.043	−0.058	0.04
CW_Marxist	0.050	0.045	0.039	0.045	0.020	0.044	0.008	0.04
COL_British	−0.007	0.134	−0.001	0.137	−0.005	0.141	−0.019	0.14
COL_Portuguese	−0.203	0.195	−0.188	0.199	−0.175	0.205	−0.178	0.20
COL_Spanish	0.135	0.356	0.112	0.364	0.081	0.375	0.063	0.37
COL_Italian	−0.448+	0.267	−0.490+	0.272	−0.518+	0.280	−0.517+	0.28
COL_Belgian	0.180	0.356	0.188	0.363	0.195	0.375	0.176	0.37
COL_German	−0.318*	0.160	−0.333*	0.164	−0.347*	0.169	−0.351*	0.17
COL_US	0.118	0.331	0.115	0.338	0.105	0.349	0.102	0.35
COL_None	0.160	0.361	0.162	0.368	0.151	0.379	0.122	0.38
REG_E_Africa	−0.324+	0.195	−0.349+	0.199	−0.368+	0.205	−0.363+	0.20
REG_S_Africa	−0.670**	0.211	−0.695**	0.215	−0.719**	0.222	−0.712**	0.22
REG_C_Africa	−0.189	0.214	−0.205	0.219	−0.230	0.225	−0.227	0.22
REG_W_Africa	−0.543**	0.184	−0.559**	0.187	−0.582**	0.193	−0.572**	0.19
SD (Intercept YEAR)	0.366		0.356		0.338		0.337	
SD (Intercept CCODE)	0.310		0.318		0.328		0.330	
SD (Observations)	0.327		0.321		0.316		0.312	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 18: **HHI Total: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−9.732***	1.960	−10.409***	2.000	−10.706***	2.036	−12.059***	2.109
TOTLIB1	0.008+	0.004						
TOTLIB1_lag1			0.012**	0.004				
TOTLIB1_lag2					0.014**	0.004		
TOTLIB1_lag3							0.015***	0.004
logPOP	7.891***	1.676	8.368***	1.706	8.381***	1.738	9.504***	1.797
logPOP^2	−2.011***	0.455	−2.175***	0.463	−2.228***	0.472	−2.533***	0.487
logPOP^3	0.150***	0.041	0.166***	0.042	0.174***	0.043	0.200***	0.044
logGNI	0.001	0.174	0.200	0.178	0.416*	0.181	0.545**	0.186
logGNI^2	0.022	0.024	−0.007	0.025	−0.036	0.025	−0.055*	0.026
logTRDIMtot	0.037	0.029	0.043	0.029	0.045	0.029	0.026	0.030
logTRDEXtot	−0.026	0.028	−0.022	0.028	−0.020	0.028	0.000	0.028
COLD_WAR	0.120*	0.058	0.064	0.047	0.046	0.045	0.012	0.040
CW_Socialist	−0.081	0.053	−0.082	0.052	−0.080	0.053	−0.075	0.053
CW_Marxist	−0.008	0.047	−0.003	0.047	0.001	0.047	0.010	0.047
COL_British	0.032	0.163	0.048	0.168	0.049	0.168	0.062	0.175
COL_Portuguese	−0.225	0.222	−0.225	0.229	−0.227	0.229	−0.201	0.238
COL_Spanish	−0.306	0.431	−0.325	0.444	−0.320	0.444	−0.374	0.462
COL_Italian	−0.507	0.325	−0.531	0.334	−0.499	0.334	−0.527	0.347
COL_Belgian	0.125	0.439	0.153	0.452	0.120	0.452	0.141	0.471
COL_German	−0.322	0.199	−0.319	0.205	−0.324	0.205	−0.319	0.213
COL_US	0.083	0.412	0.065	0.425	0.072	0.425	0.055	0.443
COL_None	0.141	0.444	0.174	0.457	0.160	0.457	0.179	0.476
REG_E_Africa	−0.412+	0.240	−0.471+	0.246	−0.493*	0.246	−0.517*	0.256
REG_S_Africa	−0.810**	0.255	−0.865***	0.262	−0.878***	0.263	−0.937***	0.273
REG_C_Africa	−0.249	0.259	−0.304	0.266	−0.314	0.266	−0.363	0.276
REG_W_Africa	−0.670**	0.224	−0.716**	0.230	−0.724**	0.230	−0.768**	0.239
SD (Intercept YEAR)	0.154		0.114		0.106		0.087	
SD (Intercept CCODE)	0.388		0.400		0.401		0.418	
SD (Observations)	0.310		0.306		0.303		0.301	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001